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SUPERSEDING
MIL-M-6270B(USAF)
5 May 1969

MILITARY SPECIFICATION

MITTEN INSERTS, AIRCREW, WOOL, HEAVY, TYPE N-3B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of heavy, knitted wool mittens, furnished in one size, designated type N-3B.

2. APPLICABLE DOCUMENTS

*2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

*2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	Thread, Polyester
UU-P-268	Paper, Kraft, Wrapping
DDD-L-20	Label, For Clothing, Equipage, And Tentage, (General Use)
PPP-B-636	Box, Shipping Fiberboard
PPP-F-320	Fiberboard, Corrugated And Solid Sheet, Sheet Stock (Container Grade), And Cut Shapes

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, And Stitchings

Beneficial comments (recommendations, additions,deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer)

*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82	Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

U S. Department of Agriculture Methods of Test for Grades of Wool

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington DC 20250.)

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Guide samples when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.1.2 Standard sample. The finished mitten shall match the standard sample for shade and shall be equal to or better than the standard sample, with respect to all the characteristics for which the standard sample is referenced (see 6.3).

3.1.3 First article. When specified (see 6.2), before production is commenced, a first article sample consisting of one dozen pairs of finished mittens shall be submitted to the Contracting Officer or his authorized representative for inspection as specified in 4.3.2 and 4.3.3. The approval of the first article sample authorizes the commencement of production but

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does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article sample shall be manufactured by the supplier in the same facilities to be used for manufacture of the production items.

3.2 Materials.

3.2.1 Wool. The wool used in the manufacture of the mittens shall be sound, strong staple, fleece or pulled wool, not lower in grade than 56's current U.S. Standard. Noils, card fly, card stripping, reused wool, reprocessed wool fiber obtained from sweater clips, garnetted hard ends, and similar waste shall not be used. The wool shall not contain vegetable matter. The contractor shall certify compliance with all requirements of this paragraph except the wool grade which shall be tested in accordance with 4.3.1.1 or 4.3.1.2.

3.2.2 Yarn. The yarn shall be manufactured on either the woolen or the worsted system. The wool shall be evenly carded, spun, and twisted for the woolen yarn, or shall be evenly carded, combed, drawn, spun, and twisted for the worsted yarn into the following sizes (plus or minus tolerance of 5 percent).

	Woolen Spun	Worsted Spun
Cuff and Hand	2/14's cut	2/7-1/2's
Thumb	3/14's cut	3/7-1/2's

3.2.3 Thread. The thread for all sewing shall be size E, conforming to V-T-285.

3.3 Color. The color of the knit wool shall be Sage Green, USAF Shade 1525 (see 6.3), the color of the thread shall match the color of the mittens knit wool.

3.3.1 Matching. The color of the dyed mitten shall match the standard sample under natural (north sky) daylight, or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4 Dimensions. The finished mittens shall conform to the measurements specified in Table I and Figure 1.

3.5 Construction.

3.5.1 The mittens shall be constructed in accordance with 3.5.1.1, 3.5.1.2, 3.5.1.3 and 3.5.1.4 and applicable stitches, seams, and stitching of FED-STD-751.

3.5.1 1 Hand. The hand shall be seamless plain stitch, knitted on a 4 to 4-3/4 inch diameter knitting machine, having not less than 64 needles. Two ends of yarn shall enter each feed.

3.5.1.2 Hand seaming. The top of the hand shall be stitched on a smooth curve as shown in Figure 1 and shall be closed by seaming on the inside with stitch Type 401, seam Type SSa-1, having not less than ten stitches per inch. The thumb may be machine closed with 401 stitch, seam SSa-1, ten to 14 stitches per inch using polyester thread.

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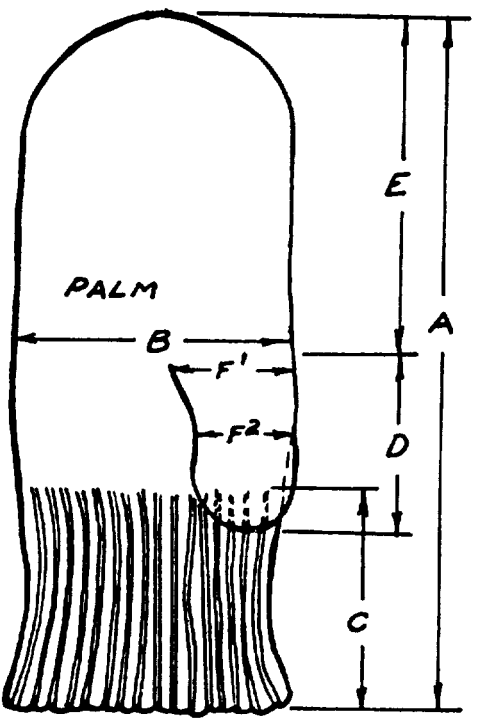


FIGURE 1 TYPE N-3B MITTEN

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TABLE I. Mittens Measurements.

Location	Symbol (Figure 1)	Measurement (Inches)
Length	A	12
Width at Thumb	B	4-5/8
Cuff Length	C	3-1/2
Length of Thumb	D	3
Thumb Crotch to Top	E	5-3/4
Width of Thumb	F[1] *F[2]	1-7/8 1-1/2

Tolerance: Plus or minus 1/4 inch.

*Measurement at location which is 1/2 the length of thumb.

3.5.1.3 Cuff. The cuff shall be rib knitted, one by one rib. It shall be knitted on a 4 to 4-3/4 inch diameter cylinder knitting machine having not less than 64 needles, then knitted or transferred to the hand. Two ends shall enter each feed. The top edge of the cuff shall be knitted with a nonraveling welt. The seam shall be closed with a Type 301 stitch with not less than twelve stitches per inch.

3.5.1.4 Thumb. The thumb shall be knitted seamless and shall be inserted on a direct line with the edge of the hand to prevent undue strain. The bottom shall be knitted to an opening of 28 needles and narrowed down four needles and hand finished with the same yarn.

3.5.1.5 Fulling and finishing. The mittens shall be thoroughly scoured, fulled, rinsed clean and, with the exception of the thumb, brushed or napped on the inside after fulling. The finished mittens shall be formed to proper shape and size, shall be flexible, and the outside shall have a minimum of protruding wool fibers.

3.5.1.6 Pairing. The mittens shall be paired and tacked together.

3.6 Weight. The weight of one dozen pairs of mittens shall be not less than 50 ounces.

3.7 Marking.

3.7.1 Label. Each mitten in each pair shall be marked with identification label conforming to Type III or IV, class 4 of DDD-L-20 except that the color of the marking shall be white. The label shall be applied to the palm side, of the inside of the cuff, 1/2 inch from the top edge. The markings shall be clearly legible and show good fastness to laundering when tested as specified in DDD-L-20. The label legend shall be as listed below, and the size of characters shall be a minimum of 1/4 inch.

Nomenclature
Specification MIL-M-6270C(USAF)

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Contract Number
Federal Stock Number
Product Labeling Act Information
Contractor Name and Address

3.7.2 Wool products label. Mittens shall be marked so as to conform to the Wool Products Labeling Act of 1939.

3.8 Workmanship. The finished mittens shall conform to the quality and grade of products established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article sample inspection. When required, inspection and testing of the first article sample shall be made of the finished mittens for all provisions of this specification applicable to end product examination and tests.

4.2.1 Certificate of compliance. When certificate of compliance is submitted, the Government reserves the right to check test such items to determine the validity of the certificate.

4.3 Sampling for inspection. Sampling for inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed below for the characteristics shown. The method of testing specified in FED-STD-191 when applicable and as specified below shall be followed.

4.3.1.1 Wool grade (grease or scoured wool). The grade of wool when made in grease or scoured form shall be determined by visual comparison with the applicable U.S. Standard prior to dyeing, picking and carding. In the event of dispute resulting from the above comparison, the wool grade shall be determined by a width method (wedge) approved by the Government. The examination for grade shall be performed on a composite sample of 1 pound for each 10,000 pounds or fraction thereof in the lot. The lot shall be unacceptable if the sample does not conform to the requirements of 3.3.1.

4.3.1.2 Wool grade (top). When tested in top form the grade shall be determined by comparison to the U.S. Standard. In the event of a dispute, the wool grade shall be determined by a Government approved width method. The unit of product shall be one (1) yard of undyed top. The sample size (number of units of product) shall be as shown in the table below:

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Lot Size (Pounds)	Sample Size
800 or less	2
801 to 22,000	3
22,001 and over	5

The result of each unit of product shall be reported separately. The lot shall be unacceptable if a single unit of product fails to meet the required grade.

4.3.1.3 Yarn. The yarn shall be tested in accordance with Table II. The lot size shall be expressed in units of 1 pound. The unit of product for each yarn size shall be 120 yards. The sample size shall be as follows:

Lot Size (Units)	Sample Size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE II. Component Tests

Characteristic	Reqmt Para	MIL-STD-191 Test Method	No. Det. Per Unit of Product	Result Reported As
Yarn size	3.2.2	4054	1	To the nearest 1/2 count
Ply	3.2.2	4020	1	Pass or fail

The results of each unit of product shall be reported separately. The lot shall be unacceptable if a single unit of product fails to meet the requirement for any characteristic.

4.3.2 Inspection of end item.

4.3.2.1 Visual examination. Defects found during this examination shall be classified in accordance with Table III. The lot size shall be expressed in units of one mitten. The unit of product shall be one mitten. The acceptable quality level (AQL) shall be 2.5 major defects and 6.5 total defects per hundred units. The sample size shall be level I of MIL-STD-105.The sample shall be drawn in pairs.

4.3.2.2 Examination for dimensions and pairing. The defects found during this examination shall be classified in accordance with the list below. The lot size shall be expressed in units of one mitten. The unit of product shall be one mitten. The sample size shall be in accordance with level S-3 of MIL-STD-105. The acceptable quality level shall be 4.0 defects per hundred units (one class).

Any measurement exceeding the specified plus or minus tolerances.

Mispaired, i.e., 2 mittens for the same hand (one defect per pair shall be scored).

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TABLE III Classification of defects for general visual examination

EXAMINE	DEFECT	Classification	
		MAJOR	MINOR
Material	Hole, run or dropped stitch	X	
	Cut or tear	X	
	Mend	X	
	Thin or uneven fabric	X	
	Shives or burrs	X	
	Slubby yarn more than twice the thickness of the normal yarn		X
	Spot, stain or streak		X
	Knot, large oversized protruding from the surface		X
	Welt, frayed or raveled		X
	Floated ends more than 1 inch in length		X
Off shade	Shade not comparable to standard sample or approved sample (off shade), as a minor defect		X
Seams and stitching	Incorrect seam or stitch type		X
	Less than specified stitches per inch		X
	Open seams, broken or skipped stitches	X	
	Loose or tight tension affecting appearance		X
	Seam, twisted, pleated or puckered		X
Construction	Thumb improperly positioned	X	
	Thumb tip open or insecurely closed	X	
	Any portion of mitten poorly shaped		X
	More than 3 untrimmed ends over 1 inch in length		X
Labels	Contract or wool products label missing	X	
	Misplaced, incomplete, incorrect, or illegible		X
	Not attached as specified	X	

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Any of the following measurements exceeding the specified plus or minus tolerances

Location	Symbol (Figure 1)	Measurement (Inches)
Length	A	12
Width at Thumb	B	4-5/8
Cuff Length	C	3-1/2
Length of Thumb	D	3
Thumb Crotch to Top	E	5-3/4
Width of Thumb	F^1	1-7/8
Width of Thumb	F^2	1-1/2

4.3 2.3 Examination for preparation of delivery. Examination shall be made to determine that the packaging, packing and marking as required by Section 5 of this specification are complied with. The lot size shall be the number of containers in the inspection lot The unit of product shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure shall be examined on shipping containers fully prepared for delivery.

Lot size	Sample size
Up to 1200	3
1201 up	5

The lot shall be unacceptable if one of the following defects is found in the sample.

Examination	Defect
Markings Exterior and Interior	Incorrect, missing, illegible, of improper size, location, sequence or method of application. Size on item marking not in conformance with size shown on interior or exterior package.
Materials	Any nonconforming component, component missing damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps loose strapping, etc Bulging or distortion of containers.
Weight (when applicable)	Gross or net weight more than specified

4.3.2.4 Examination for count. Table IV shall govern this examination.

TABLE IV. Examination for count.

EXAMINATION	LOTS EXPRESSED IN UNITS OF	UNIT OF PRODUCT	SAMPLE SIZE	
			NUMBER OF UNITS IN LOT	NUMBER OF UNITS SELECTED
a. Number of pairs per intermediate package	One intermediate packaging	One intermediate package	N/A	Same as for Visual examination
b. Number of pairs of mittens per shipping Container 1/	One shipping container	One shipping container	Up to 3 4 to 6 7 and over	All 3 4

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1/ Count intermediate packages and multiply by the pairs of mittens per intermediate package.

The lot shall be unacceptable if any of the following are found:

a. Number of pairs of mittens per intermediate package is less than specified.
b. Number of pairs of mittens per shipping container is less than specified or less than the quantity marked on the container.

4.3.3 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable and as specified in Table V shall be followed. The physical and chemical values specified apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of one mitten. The unit of product shall be as follows:

For weight - One dozen pairs with a sample size of one regardless of lot size.

For all other tests - One mitten.

TABLE V. Testing of end item.

CHARACTERISTICS	REFERENCE PARAGRAPH	TEST METHOD	# OF DET. PER UNIT OF PRODUCT	RESULTS REPORTED AS
Color	3.3	Visual	1	Pass or fail
Number of needles	3.5.1.1 3.5.1.3 3.5.1.4	count per inch	1 in each area	To the nearest needle
Weight	3.6	1/	1	To the nearest ounce per dozen

The sample size shall be as specified in 4.3.1.3 and the lot shall be unacceptable if one or more units fail to meet any requirements specified.

* 1/ The twelve mittens encompassing the sample unit for testing shall be allowed to reach equilibrium under standard conditions as defined in FED-STD-191. When under standard conditions, the mittens shall be weighed as a unit to the nearest 1.0 ounce.

*5. PACKAGING

*5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

*5.1.1 Level A. Each pair of mittens shall be tacked together with light thread. Five (5) pairs of mittens shall be neatly stacked in a bundle and secured at end with cotton tape or twine.

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*5.1.2 Commercial. Each pair of mittens shall be packaged in accordance with the requirements of ASTM D 3951.

*5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A. Sixty (60) pairs (twelve bundles) of mittens packaged as specified shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, class weather-resistant type SF, grade V2s of PPP-B-636. The fiberboard for the liner shall conform to type CF, class weather-resistant variety SW, grade V3c of PPP-F-320. Three ounces of flaked naphthalene shall be uniformly distributed throughout the contents of each shipping container. Each container shall have the contents completely covered on top and bottom with a sheet of 30 pound minimum basic weight kraft paper conforming to grade B of UU-P-268. Each container shall be closed and water-proofed in conformance with the requirements of the appendix of the box specification.

5.2.2 Level B. Sixty pairs (twelve bundles) of mittens packaged as specified shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, class domestic, type CF or SF, grade 275 of PPP-B-636. The fiberboard for the liner shall conform to type CF, class domestic, variety SW, grade 275 of PPP-F-320. Three ounces of flaked naphthalene shall be uniformly distributed throughout the contents of each shipping container. Each containter shall have the contents completely covered on top and bottom with a sheet of 30 pound minimum basic weight kraft paper conforming to grade B of UU-P-268. Each container shall be closed in conformance with Method II of the appendix of the box specification.

5.2.3 Commercial. Sixty (60) pairs (twelve bundles) mittens packaged as specified in 5.1 shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The wool mitten covered by this specification are intended to be worn by aircrew personnel underneath leather mittens.

6.2 Ordering data.

*6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When first article sample is not required (see 3.1).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Samples. For access to standard shade sample, address the procuring office issuing the invitation for bids.

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

Review Activity:

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Air Force 45

Preparing Activity:

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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